**1. Sheffer stroke (9 points)**

Is the Sheffer stroke | associative? Show this by proving the semantic equivalence $(\phi \mid \psi) \mid \chi \equiv \phi \mid (\psi \mid \chi)$, or bring a counterexample to disprove it.

SOLUTION

| $\phi$ | $\psi$ | $\chi$ | $(\phi \mid \psi)$ | $(\psi \mid \chi)$ | $(\phi \mid \psi) \mid \chi$ | $\phi \mid (\psi \mid \chi)$, |
|---|---|---|---|---|---|---|
| T | T | T | F | F | T | T |
| T | T | F | F | T | T | F |
| T | F | T | T | T | F | F |
| T | F | F | T | T | T | T |
| F | T | T | T | F | F | T |
| F | T | F | T | T | T | T |
| F | F | T | T | T | F | T |
| F | F | F | T | T | T | T |

As the last two columns of the truth table do not match on every row, the Sheffer stroke is not associative. A counterexample is when $\phi$ and $\psi$ are true and $\chi$ is false: in this case, $(\phi \mid \psi) \mid \chi$ is true, but $\phi \mid (\psi \mid \chi)$ is false.

**2. Island puzzle (12 points)**

On the island of liars and truth speakers, every resident either always speaks the truth or always lies. Suppose that you meet three islanders, A, B and C. A says *'B is a liar'*, B says *'C is a truth speaker'*, and C says *'A and B are both liars'*. Determine, via a truth table, whether A, B and C are liars or truth speakers.

SOLUTION:

Let us use three propositional variables:

$t_A$: A is a truth speaker
$t_B$: B is a truth speaker
$t_C$: C is a truth speaker

We can construct the following truth table:

| $t_A$ | $t_B$ | $t_C$ | $t_A \leftrightarrow \neg t_B$ | $t_B \leftrightarrow t_C$ | $t_C \leftrightarrow \neg t_A \wedge \neg t_B$ |
|---|---|---|---|---|---|
| T | T | T | F | T | F |
| T | T | F | F | F | T |
| T | F | T | T | F | F |
| T | F | F | T | T | T |
| F | T | T | T | T | F |
| F | T | F | T | F | T |
| F | F | T | F | F | T |
| F | F | F | F | T | F |

The highlighted row is the only one which makes all three bi-implications true, which means it is the only one which can represent reality. Therefore, A must be a truth speaker, while B and C must be liars.

**3. CNF algorithm (12 points)**

Show, using the CNF algorithm, that the formula $(p \wedge \neg q) \rightarrow \neg(q \rightarrow p)$ is semantically equivalent to $\neg p \vee q$.

SOLUTION:

1. Implication-free:
   $\neg(p \wedge \neg q) \vee \neg(q \rightarrow p)$
2. Implication-free:
   $\neg(p \wedge \neg q) \vee \neg(\neg q \vee p)$
3. Bringing negations inside brackets:
   $(\neg p \vee q) \vee (q \wedge \neg p)$
4. Distributivity:
   $(\neg p \vee q \vee q) \wedge (\neg p \vee q \vee \neg p)$
5. Simplifying the solution:
   $\neg p \vee q$

**4. DPLL procedure (12 points)**

Apply the DPLL procedure to the CNF $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg q \vee r)$ to show whether it is satisfiable. Explain in detail how the procedure comes to a result. Unless there is unit clause or a pure literal, let the procedure branch first on *p*, then on *q*, and then on *r*; and let it always try the truth value true (T) before false (F).

SOLUTION

1. No unit clauses to eliminate.
2. Pure literal: r. Let us try $r \equiv T$:
   $(p \vee \neg q) \wedge (\neg p \vee q) \wedge T \equiv (p \vee \neg q) \wedge (\neg p \vee q)$
3. Let us try $p \equiv T$:
   $(T \vee \neg q) \wedge (\bot \vee q) \equiv q$
4. Let then be $q \equiv T$: the formula becomes true. Therefore, it is satisfiable, and a model is when p, q and r are all true.

**5. Sets (6 + 9 points)**

Consider the following set-theoretic equality:

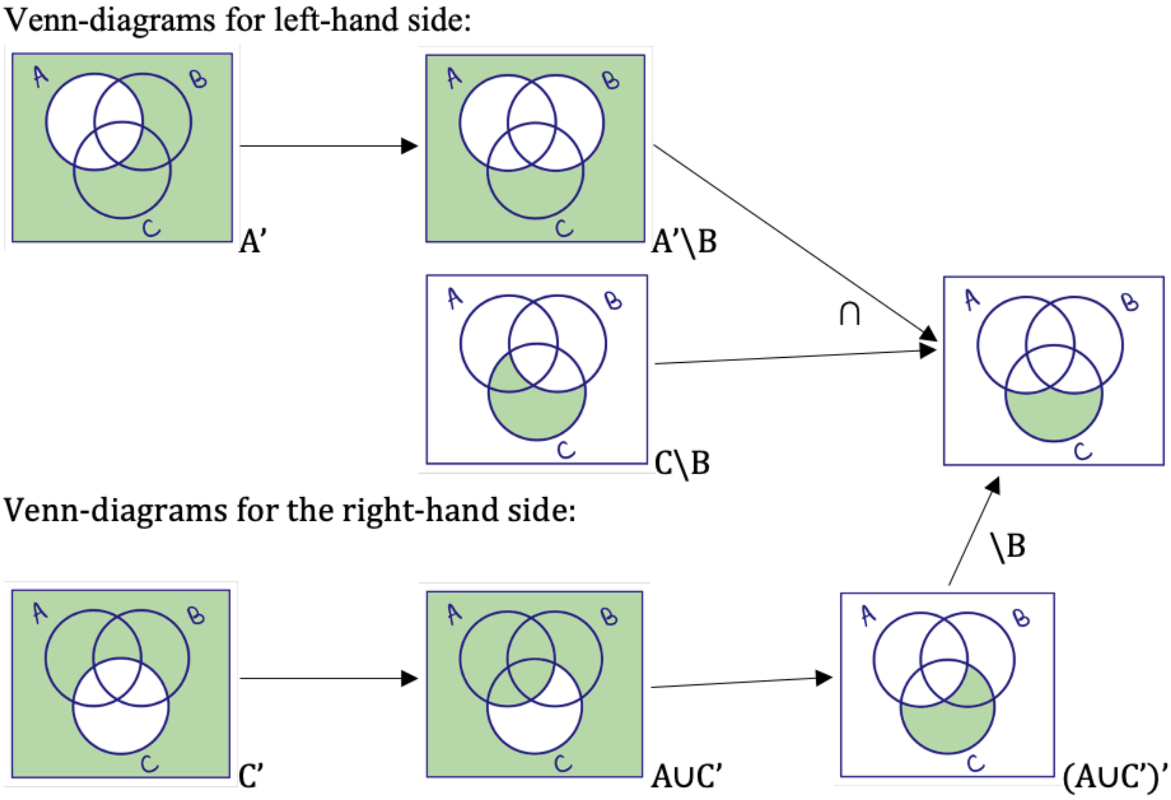
$$(A' \setminus B) \cap (C \setminus B) = (A \cup C')' \setminus B$$

(a) Prove this identity using a sequence of Venn diagrams. For each side of the equality, you can draw a sequence of diagrams which shows how the set is constructed from A, B, and C via elementary set operations, and use this to prove that the equation holds for any three sets A, B and C.

(b) Prove this identity using the rules of the algebra of sets. Please make sure to name the rules which you apply.

SOLUTION

(a)




(b)

Left-hand side:

| | | | |
|---|---|---|---|
| $(A' \setminus B) \cap (C \setminus B)$ | = | $(A' \cap B') \cap (C \cap B')$ | Definition of difference |
| | | $A' \cap B' \cap C \cap B'$ | Associativity |
| | | $A' \cap C \cap B' \cap B'$ | Commutativity |
| | | $A' \cap C \cap (B' \cap B')$ | Associativity |
| | | $A' \cap C \cap B'$ | Idempotence |
| | | $A' \cap B' \cap C$ | Commutativity |

Right-hand side:

| | | | |
|---|---|---|---|
| $(A \cup C')' \setminus B$ | = | $(A' \cap (C')') \setminus B$ | De Morgan rule |
| | | $(A' \cap C) \setminus B$ | Involution |
| | | $(A' \cap C) \cap B'$ | Definition of difference |
| | | $A' \cap C \cap B'$ | Associativity |
| | | $A' \cap B' \cap C$ | Commutativity |

The derivations above show that both sides are equivalent to $A' \cap B' \cap C$, and so equivalent to each other.

**6. Relations (5 + 6 + 6 points)**

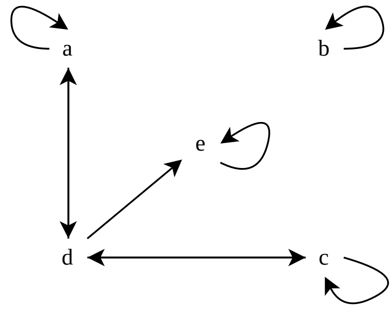
Consider the set V = {a, b, c, d, e} and W = {1, 2, 3, 4, 5} and the relations:

$$R = \{(a, a), (a, d), (b, b), (c, c), (c, d), (d, a), (d, c), (d, e), (e, e)\}$$
$$S = \{(a, 2), (b, 3), (d, 3), (d, 5), (e, 4)\}$$

a) Draw the directed graph representation of the relation R.
b) Explain whether the relation R is transitive and/or anti-symmetric.
c) Write down the matrix representation of $S \circ R$. It might help to illustrate the relations in a Venn-diagram first.

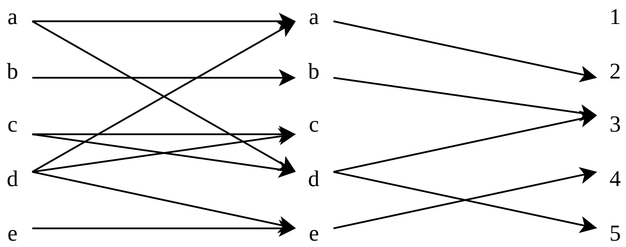
SOLUTION:

(a)




(b) It is not transitive as it has (c, d) and (d, e) but not (c, e). It is also not anti-symmetric as we have (a, d) and (d, a) as well as (c, d) and (d, c).

(c)




| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| a | 0 | 1 | 1 | 0 | 1 |
| b | 0 | 0 | 1 | 0 | 0 |
| c | 0 | 0 | 1 | 0 | 1 |
| d | 0 | 1 | 0 | 1 | 0 |
| e | 0 | 0 | 0 | 1 | 0 |

**7. Ordering relations (7 + 4 points)**

Consider the sets $X \coloneqq \{x_1,x_2,x_3\}$ and $Y \coloneqq \{y_1,y_2,y_3\}$, with partial orders $\leq_X$ and $\leq_Y$ on these respective sets described by the following Hasse diagrams:

$$\begin{tikzcd}[column sep=small] & x2 & & & y2 \arrow[dl] \arrow[dr] & \\ x1 \arrow[ur] & & x3 \arrow[ul] & y1 & & y3 \end{tikzcd}$$

(a) Draw the Hasse diagram of the Cartesian ordering relation on X × Y induced by $\leq_X$ and $\leq_Y$ by adding arrows to the picture below. You can use the algorithm you have learned about in class, but you are not required to do so.

$$\begin{array}{ccc} \langle x_1, y_1 \rangle & \langle x_1, y_2 \rangle & \langle x_1, y_3 \rangle \\ \\ \langle x_2, y_1 \rangle & \langle x_2, y_2 \rangle & \langle x_2, y_3 \rangle \\ \\ \langle x_3, y_1 \rangle & \langle x_3, y_2 \rangle & \langle x_3, y_3 \rangle \end{array}$$

(b) Determine and list all the maximal and all the minimal elements of X × Y in this Cartesian ordering. Is there a maximum? Is there a minimum?

SOLUTION

(a)

$$\begin{tikzcd} \langle x_1, y_1 \rangle \arrow[d] & \langle x_1, y_2 \rangle \arrow[l] \arrow[r] \arrow[d] & \langle x_1, y_3 \rangle \arrow[d] \\ \langle x_2, y_1 \rangle & \langle x_2, y_2 \rangle \arrow[l] \arrow[r] & \langle x_2, y_3 \rangle \\ \langle x_3, y_1 \rangle \arrow[u] & \langle x_3, y_2 \rangle \arrow[l] \arrow[r] \arrow[u] & \langle x_3, y_3 \rangle \arrow[u] \end{tikzcd}$$

(b) There is not a maximum, but there are two maximal elements, $< x_2, y_1>$ and $< x_2, y_3>$. Similarly, there is not a minimum but there are two minimal elements, $<x_1, y_2>$ and $<x_3, y_2>$.